Module 5

Exercise 3 (Postgress)

Part 1: Comparison and Logical Operators

1. Retrieve all products with a unit price greater than 20.
A.) SQL

```
select product_id,product_name,unit_price
from products
where unit_price>20;
```

Output

product_id	product_name	unit_price
4	Chef Anton's Cajun Seasoning	22
5	Chef Anton's Gumbo Mix	21.35
6	Grandma's Boysenberry Spread	25
7	Uncle Bob's Organic Dried Pears	30
8	Northwoods Cranberry Sauce	40
9	Mishi Kobe Niku	97
10	Ikura	31
11	Queso Cabrales	21
12	Queso Manchego La Pastora	38
14	Tofu	23.25
17	Alice Mutton	39
18	Carnarvon Tigers	62.5
20	Sir Rodney's Marmalade	81
22	Gustaf's Knäckebröd	21
26	Gumbär Gummibärchen	31.23
27	Schoggi Schokolade	43.9
28	Rössle Sauerkraut	45.6
29	Thüringer Rostbratwurst	123.79
30	Nord-Ost Matjeshering	25.89
32	Mascarpone Fabioli	32
37	Gravad lax	26
38	Côte de Blaye	263.5
43	Ipoh Coffee	46
51	Manjimup Dried Apples	53
53	Perth Pasties	32.8
55	Pâté chinois	24
56	Gnocchi di nonna Alice	38
59	Raclette Courdavault	55
60	Camembert Pierrot	34
61	Sirop d'érable	28.5
62	Tarte au sucre	49.3
63	Vegie-spread	43.9
64	Wimmers gute Semmelknödel	33.25
65	Louisiana Fiery Hot Pepper Sauce	21.05
69	Gudbrandsdalsost	36
71	Flotemysost	21.5
72	Mozzarella di Giovanni	34.8

(37 rows)

2. Fetch all employees whose birth date is before '1960-01-01'.
A.)SQL

```sql
select employee_id,first_name,last_name,birth_date
from employees
where birthdate<'1960-01-01';
```

Output

```
 employee_id | first_name | last_name | birth_date
-------------+------------+-----------+------------
           1 | Nancy      | Davolio   | 1948-12-08
           2 | Andrew     | Fuller    | 1952-02-19
           4 | Margaret   | Peacock   | 1937-09-19
           5 | Steven     | Buchanan  | 1955-03-04
           8 | Laura      | Callahan  | 1958-01-09
(5 rows)
```

3. Get a list of customers who are from either Germany, France, or the UK using the IN operator.

A.)SQL

```sql
select customer_id,company_name,contact_name,country
from customers
where country IN ('Germany','France','UK');
```

Output

customer_id	company_name	contact_name	country
ALFKI	Alfreds Futterkiste	Maria Anders	Germany
AROUT	Around the Horn	Thomas Hardy	UK
BLAUS	Blauer See Delikatessen	Hanna Moos	Germany
BLONP	Blondesddsl père et fils	Frédérique Citeaux	France
BONAP	Bon app'	Laurence Lebihan	France
BSBEV	B's Beverages	Victoria Ashworth	UK
CONSH	Consolidated Holdings	Elizabeth Brown	UK
DRACD	Drachenblut Delikatessen	Sven Ottlieb	Germany
DUMON	Du monde entier	Janine Labrune	France
EASTC	Eastern Connection	Ann Devon	UK
FOLIG	Folies gourmandes	Martine Rancé	France
FRANK	Frankenversand	Peter Franken	Germany
FRANR	France restauration	Carine Schmitt	France
ISLAT	Island Trading	Helen Bennett	UK
KOENE	Königlich Essen	Philip Cramer	Germany
LACOR	La corne d'abondance	Daniel Tonini	France
LAMAI	La maison d'Asie	Annette Roulet	France
LEHMS	Lehmanns Marktstand	Renate Messner	Germany
MORGK	Morgenstern Gesundkost	Alexander Feuer	Germany
NORTS	North/South	Simon Crowther	UK
OTTIK	Ottilies Käseladen	Henriette Pfalzheim	Germany
PARIS	Paris spécialités	Marie Bertrand	France
QUICK	QUICK-Stop	Horst Kloss	Germany
SEVES	Seven Seas Imports	Hari Kumar	UK
SPECD	Spécialités du monde	Dominique Perrier	France
TOMSP	Toms Spezialitäten	Karin Josephs	Germany
VICTE	Victuailles en stock	Mary Saveley	France
VINET	Vins et alcools Chevalier	Paul Henriot	France
WANDK	Die Wandernde Kuh	Rita Müller	Germany

(29 rows)

4. Find all orders placed in the year 1997.

A.)SQL

```sql
select order_id,order_date,employee_id
from orders
where order_date BETWEEN '1997-01-01' AND '1997-12-31';
```

Output

10782	1997-12-17	9
10783	1997-12-18	4
10784	1997-12-18	4
10785	1997-12-18	1
10786	1997-12-19	8
10787	1997-12-19	2
10788	1997-12-22	1
10789	1997-12-22	1
10790	1997-12-22	6
10791	1997-12-23	6
10792	1997-12-23	1
10793	1997-12-24	3
10794	1997-12-24	6
10795	1997-12-24	8
10796	1997-12-25	3
10797	1997-12-25	7
10798	1997-12-26	2
10799	1997-12-26	9
10800	1997-12-26	1
10801	1997-12-29	4
10802	1997-12-29	4
10803	1997-12-30	4
10804	1997-12-30	6
10805	1997-12-30	2
10806	1997-12-31	3
10807	1997-12-31	4

(408 rows)

5. Display all products where the discontinued status is false.

A.)SQL

```sql
select product_id,product_name,discontinued
from products
where discontinued=0;
```

Output

```
          46 | Spegesild                          |              0
          47 | Zaanse koeken                      |              0
          48 | Chocolade                          |              0
          49 | Maxilaku                           |              0
          50 | Valkoinen suklaa                   |              0
          51 | Manjimup Dried Apples              |              0
          52 | Filo Mix                           |              0
          54 | Tourtière                          |              0
          55 | Pâté chinois                       |              0
          56 | Gnocchi di nonna Alice             |              0
          57 | Ravioli Angelo                     |              0
          58 | Escargots de Bourgogne             |              0
          59 | Raclette Courdavault               |              0
          60 | Camembert Pierrot                  |              0
          61 | Sirop d'érable                     |              0
          62 | Tarte au sucre                     |              0
          63 | Vegie-spread                       |              0
          64 | Wimmers gute Semmelknödel          |              0
          65 | Louisiana Fiery Hot Pepper Sauce |              0
          66 | Louisiana Hot Spiced Okra          |              0
          67 | Laughing Lumberjack Lager          |              0
          68 | Scottish Longbreads                |              0
          69 | Gudbrandsdalsost                   |              0
          70 | Outback Lager                      |              0
          71 | Flotemysost                        |              0
          72 | Mozzarella di Giovanni             |              0
          73 | Röd Kaviar                         |              0
          74 | Longlife Tofu                      |              0
          75 | Rhönbräu Klosterbier               |              0
          76 | Lakkalikööri                       |              0
          77 | Original Frankfurter grüne Soße    |              0
(67 rows)
```

Part 2: Arithmetic and String Operators

1. Show a list of products with a new column that displays unit price increased by 15%.

A.)SQL

AI Assistant Query Query History

```
SELECT product_name,unit_price,unit_price * 1.15 AS increased_price
FROM products;
```

Output

Manjimup Dried Apples	53	60.949999999999996
Filo Mix	7	8.049999999999999
Perth Pasties	32.8	37.719999122619626
Tourtière	7.45	8.567499780654906
Pâté chinois	24	27.599999999999998
Gnocchi di nonna Alice	38	43.699999999999996
Ravioli Angelo	19.5	22.424999999999997
Escargots de Bourgogne	13.25	15.237499999999999
Raclette Courdavault	55	63.24999999999999
Camembert Pierrot	34	39.099999999999994
Sirop d'érable	28.5	32.775
Tarte au sucre	49.3	56.69499912261963
Vegie-spread	43.9	50.48500175476074
Wimmers gute Semmelknödel	33.25	38.2375
Louisiana Fiery Hot Pepper Sauce	21.05	24.207499122619627
Louisiana Hot Spiced Okra	17	19.549999999999997
Laughing Lumberjack Lager	14	16.099999999999998
Scottish Longbreads	12.5	14.374999999999998
Gudbrandsdalsost	36	41.4
Outback Lager	15	17.25
Flotemysost	21.5	24.724999999999998
Mozzarella di Giovanni	34.8	40.01999912261962
Röd Kaviar	15	17.25
Longlife Tofu	10	11.5
Rhönbräu Klosterbier	7.75	8.9125
Lakkalikööri	18	20.7
Original Frankfurter grüne Soße	13	14.95

(77 rows)

2. Retrieve the full name of employees by concatenating FirstName and LastName.

A.)SQL

AI Assistant Query Query History

```sql
SELECT first_name || ' ' || last_name AS full_name
FROM employees;
```

Output

```
    full_name
------------------
 Nancy Davolio
 Andrew Fuller
 Janet Leverling
 Margaret Peacock
 Steven Buchanan
 Michael Suyama
 Robert King
 Laura Callahan
 Anne Dodsworth
(9 rows)
```

3. Find all products where the product name contains the letter 'a' at least twice.

A.)SQL

AI Assistant Query Query History

```sql
SELECT product_name
FROM products
WHERE product_name LIKE '%a%';
```

Output

```
Boston Crab Meat
Jack's New England Clam Chowder
Singaporean Hokkien Fried Mee
Gula Malacca
Zaanse koeken
Chocolade
Maxilaku
Valkoinen suklaa
Manjimup Dried Apples
Perth Pasties
Gnocchi di nonna Alice
Ravioli Angelo
Escargots de Bourgogne
Raclette Courdavault
Camembert Pierrot
Sirop d'érable
Tarte au sucre
Vegie-spread
Louisiana Fiery Hot Pepper Sauce
Louisiana Hot Spiced Okra
Laughing Lumberjack Lager
Scottish Longbreads
Gudbrandsdalsost
Outback Lager
Mozzarella di Giovanni
Röd Kaviar
Lakkalikööri
Original Frankfurter grüne Soße
(55 rows)
```

4. Calculate the total revenue (UnitPrice * Quantity) for each order in the OrderDetails table.

A.)SQL

AI Assistant Query Query History

```sql
SELECT order_id,SUM(unit_price * quantity) AS total_revenue
FROM order_details
GROUP BY order_id;
```

Output

order_id	total_revenue
11038	750.9999980926514
10782	12.5
10725	287.7999954223633
10423	1020
10518	4150.050006866455
10356	1106.3999938964844
10963	68
10596	1476.1000366210938
10282	155.40000534057617
10658	4667.999973297119
10283	1414.8000202178955
10579	317.75
10693	2333.999988555908
10896	750.5
10660	1701
10253	1444.7999839782715
10425	480
10774	875
10615	120
10514	8623.4499168396
10850	740
10276	419.99999046325684
10978	1500.7000007629395

5. Find all suppliers whose company name contains the word "food" (case insensitive)

A.)SQL

AI Assistant Query Query History

```sql
SELECT supplier_id,company_name
FROM suppliers
WHERE company_name ILIKE '%food%';
```

Output

```
 supplier_id |        company_name
-------------+----------------------------
          19 | New England Seafood Cannery
(1 row)
```

Part 3: Special Operators (IN, BETWEEN, LIKE, IS NULL, etc.)

1. Retrieve all orders where the Freight cost is between 50 and 100.5
A.)SQL

AI Assistant Query Query History

```sql
SELECT order_id,freight
FROM orders
WHERE freight BETWEEN 50 AND 100.5;
```

Output

order_id	freight
10250	65.83
10252	51.3
10253	58.17
10257	81.91
10260	55.09
10265	55.28
10268	66.29
10272	98.03
10273	76.07
10278	92.69
10283	84.81
10284	76.56
10285	76.83
10290	79.7
10304	63.79
10314	74.16
10319	64.5
10325	64.86
10326	77.92
10327	63.36

2. Find all orders where the ShipRegion is NULL.

A.)SQL

AI Assistant Query Query History

```sql
SELECT order_id,ship_region
FROM orders
WHERE ship_region IS NULL;
```

Output

order_id	ship_region
10248	
10249	
10251	
10252	
10254	
10255	
10258	
10259	
10260	
10263	
10264	
10265	
10266	
10267	
10270	
10273	
10274	